

25 – M5 Inventory, Costing, Repack & Loss

Module design for where stock is and what it cost: lots by location and batch, the append-only movement ledger, weighted-average costing per entity, availability for sellers, intra-entity transfers, counts and adjustments with value-threshold approval, repack recipes with expected yield, the write-off lifecycle with its loss categories and controls, expiry tasks, oversell exceptions, and opening balances.

Field	Value
Document	25 – M5 Inventory, Costing, Repack & Loss (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 6
Date	9 September 2026
Owning developer	Module developer (full-stack); the till track hosts count, write-off draft, repack and transfer-receipt screens in shop-management mode from this module's specification
Inputs	05 SRS (FR-INV-010–100, FR-LOSS-010–060); 03 state machines §9–11; 13 scenarios 3 and 4; 16 answers (E, F sections, B-06, R-02); 09 ADR-03, ADR-07, ADR-08; 17 v0.2.1 §7.5, ADR-38; 18 v0.2 part B (stock_lot, stock_movement, repack_recipe); 20 v0.3 M5; 21–24 contracts; 24B; 30 §5.5; 32
Depends on	M1 (locations, thresholds as role parameters), M2 (SKU flags, batch identity, conversions, alias), M4 (GRN, claim, DN, transfer request events), kernel (document base, links, attachments, audit, events, snapshot)
Depended on by	M3 (in-stock batch MRPs), M4 (availability, pick batches, lots consumed), M6 (lot snapshot, sale deductions arrive from it), M2 (assortment by lots), M8

Two invariants and one temptation. Movements are the only way stock changes, and every movement cites a document; balances are sums, materialised for speed and reconciled nightly. The temptation is to reject a sale that the records say is impossible – two offline tills selling the same last unit – and the answer is never: the sale happened, the lot goes negative, an exception is raised, and the next count fixes the record (E-09).

1. Purpose and scope

In scope (phase 1)	Out of scope
Stock lots: an entity's holding of a batch at a location, with acquisition cost and condition (GOOD, DAMAGED); balances derived from movements (E-01, E-02: one location per shop)	Shelf versus back-store sub-locations (E-02); bins and put-away in warehouses (doc 20 Appendix A)
Movement ledger: every quantity change cites a document; types RECEIPT, SALE, SALE_REVERSAL, TRANSFER_OUT/IN, REPACK_CONSUME/PRODUCE, COUNT_ADJUST, WRITE_OFF, RETURN_TO_SELLER, OPENING_BALANCE	Serialised items (B-05); cold-chain telemetry (E-10)
Weighted-average cost per entity per SKU; lot keeps acquisition cost; cost of goods sold at issue (E-03)	Per-transaction cost overrides; standard or FIFO costing
FEFO suggestion for expiry-tracked goods with manual override (E-04); pick lists for delivery notes	FEFO as the pricing rule (M3 separates them)
Availability for sellers (on hand minus quantities on issued, undispached delivery notes) (C-12)	Reservation on order acceptance (allocation is a plan, not a hold)
Counts: scheduled by the entity, cycle counting, count while trading, variance with value-threshold approval (E-05 to E-07)	ABC prioritisation (P2)
Adjustments as documents with approval bands; negative-stock and oversell exceptions (E-09)	–
Repack recipes and execution with expected yield; variance against expectation; reversal while eligible (B-06, ADR-07)	Federation repacking (P3, permission flip on the same mechanism)
Write-off lifecycle: nine loss categories, requester ≠ approver, witness ≠ requester (remote for single-staff shops), photographs, value-banded approval, loss attributed to the owner at the time (F-01 to F-05)	Disposal certificates (F-08); insurance recovery
Expiry warning tasks by window per category; markdown hand-off to M3 (F-06, F-07); loss tolerances (F-10); shrinkage reporting basis (F-09)	–
Intra-entity transfers at cost with receipt at the destination (C-10, C-11, E-01)	Cross-entity transfers (those are sales, M4)
Opening balances as two-party signed documents at onboarding (R-02)	Historical sales migration (none)
Returns to seller on approved claims; lot re-pointing on batch correction and SKU merge (M2 events)	–
Phase 2 designed as projections: stock ageing, reorder points and suggestions, ABC cycle counting (E-08, E-11)	–

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-INV-010	Balances by location and batch; movements as the only change; document citation	3.1, 3.2
FR-INV-020	Weighted-average costing per entity (E-03)	3.3
FR-INV-030	FEFO picking suggestion with override (E-04)	3.4
FR-INV-040	Counts scheduled per entity; cycle counting; trade during counts (E-05, E-06)	3.5, 4.1
FR-INV-050	Adjustment approval by value thresholds per SKU or tag (E-07)	3.5, 4.2
FR-INV-060	Negative stock permitted with acknowledgement; every occurrence on the exception report (E-09)	3.2, 6.2
FR-INV-070	Intra-entity transfers at cost, MPCS-mediated (C-10, C-11)	3.7, 4.5
FR-INV-080	Repack recipe with expected yield; variance separate from shrinkage (B-06)	3.6, 4.4
FR-INV-090	Reorder points system-calculated and overridable (P2)	3.9
FR-INV-100	Stock ageing with warnings (P2)	3.9
FR-LOSS-010	Loss categories recorded separately, never one adjustment (F-01)	3.8
FR-LOSS-020	Loss borne by the owner at the moment of loss (F-02, ADR-03)	3.8
FR-LOSS-030	Write-off: requester ≠ approver; witness distinct; photographs (F-04, F-05, N-07)	3.8, 4.3
FR-LOSS-040	Returns to seller for credit via claims (F-03)	6.7
FR-LOSS-050	Expiry alerts by window per category generating count tasks (F-06)	3.8, 4.6

Identifier	Requirement (abridged)	Design section
FR-LOSS-060	Markdown separate from loss; damaged-but-saleable as markdown (F-07, F-11)	3.8 (hand-off to M3)
R-02	Opening balances only; two-party signed documents	3.7, 4.7
ADR-38	Batch identity in M2; lot and cost here	3.1
H-02	Count entry and write-off draft offline	9.2

3. Domain model

Schema inventory: stock_lot, stock_movement, repack_recipe (doc 18 v0.2 part B) plus the tables introduced here.

3.1 Lot

Aspect	Design
Aggregate	stock_lot(location, batch, condition GOOD DAMAGED, qty_on_hand, unit_cost, received_at, last_movement_seq). A lot is one entity's holding of one batch at one location in one condition; damaged units received on a GRN form a DAMAGED lot (doc 24 §6.1), sellable only through an M3 markdown rule or written off.
Cost	unit_cost is the acquisition cost of that lot: trade price for goods bought from another entity, keyed cost for local supply, recipe-derived cost for repack outputs, restated by landed cost in phase 2 (M10).
Re-pointing	On batch.corrected.v1 lots move to the replacement batch (same qty and cost); on sku.merged.v1 lots of the alias SKU's batches remain (batch identity carries the SKU) and queries resolve through the alias.

3.2 Movement ledger and balances

Aspect	Design
Ledger	stock_movement(movement_seq per location, location, batch, condition, qty_delta, movement_type, unit_cost_at_movement, document, line, occurred_at, operator, device) (doc 18 part B). Insert-only. movement_seq is dense per location: assigned by the till for its own movements (inside the receipt bundle) and by central otherwise; the two sequences are kept apart by a source column so both stay dense.
Balances	stock_lot.qty_on_hand = Σ qty_delta for the lot (B-14); updated in the same transaction as the movement; the nightly StockReconcileJob recomputes and raises a REVIEW on drift, never silently fixes.
Negative lots	A SALE movement may take a lot below zero (offline oversell). The lot is flagged negative_since; lot.negative.v1 goes to the exception queue; the next count or receipt clears it. Availability never counts negative lots.
Batch attribution of a sale	The receipt line carries the batch the till resolved (M3 engine: scanned, picked or lowest-MRP); the deduction hits that lot. If that lot is exhausted, the deduction still hits it (negative) rather than silently moving to another batch, so the record shows what the till believed.

3.3 Costing

Aspect	Design
Method	Moving weighted average per entity per SKU (E-03, IAS 2): entity_sku_cost(entity, sku, qty_on_hand, avg_cost) updated on every receipt-type movement (RECEIPT, TRANSFER_IN at the receiving entity is at cost so unchanged, REPACK_PRODUCE, OPENING_BALANCE) as $(qty \times avg + qty_{in} \times cost_{in}) / (qty + qty_{in})$; issues (SALE, WRITE_OFF, TRANSFER_OUT within the entity unchanged, REPACK_CONSUME) carry unit_cost_at_movement = avg_cost at that moment.
Lot cost versus entity average	The lot keeps its acquisition cost for traceability and for claims; the entity average is the costing figure for cost of goods sold, valuation and write-off value. Both are stored; neither is recomputed retroactively.
Damaged units	A DAMAGED lot carries the same unit cost as its GOOD sibling; the loss is recognised when written off or when marked down (M3).

3.4 FEFO and picking

Aspect	Design
Suggestion	FifoOrder(location, sku) ranks GOOD lots by expiry then received_at; the till's lot snapshot carries the rank; warehouses get a pick_list per delivery note (M4 delivery_note.issued.v1) with suggested lots that the picker may override with a reason.
Availability	available(location, sku) = Σ GOOD lots on hand (non-negative) – quantities on ISSUED but not DISPATCHED delivery notes; Availability(entity locations, skus) is what M4 shows buyers and uses in allocation runs (DR-1).

3.5 Counts and adjustments

Aspect	Design
Count task	count_task(location, scope: full skus tags expiry-window, scheduled_for, status) created by the entity's schedule or by the expiry job; results as document CNT with lines (sku, batch, condition, expected_qty at count start, counted_qty); counting while trading is supported by snapshotting expected quantities at task start and applying movements that occur during the count to the expectation (E-06).
Variance and adjustment	Submitting a count creates an ADJ document with the variances; approval is banded by value (variance qty × entity avg cost) against variance_threshold(sku tag, band, approver permission with limits) (E-07); tolerances (F-10) below which a variance posts without escalation; approved adjustments post COUNT_ADJUST movements; the exception queue receives every count that cleared a negative lot.

3.6 Repack

Aspect	Design
Recipe	repack_recipe(owner, input sku, input qty, output sku, output qty, expected_loss_pct, status) (ADR-07, B-06); the output SKU is a distinct M2 SKU (origin REPACK_OUTPUT).
Execution	Document RPK: input lot(s), actual output qty; movements REPACK_CONSUME (input) and REPACK_PRODUCE (output at cost = consumed cost / produced qty); yield variance recorded on the document against the recipe's expectation and reported separately from shrinkage; the output batch is registered through M2 with the input batch's expiry and a local MRP if labelled.
Reversal	Permitted while the output lot is untouched (no sale, transfer or write-off against it; qty equals produced): a REVERSES-linked RPK restores the input lot; used for wrong-recipe corrections (doc 13 scenario 3).

3.7 Transfers and opening balances

Aspect	Design
Transfer (XFR)	Within one entity only. Issued at the source location on M4's transfer_request.approved.v1 or directly by the entity for warehouse-to-warehouse: TRANSFER_OUT at source at cost; in-transit until the destination confirms receipt (offline-capable on the till, LOCATION series) posting TRANSFER_IN; unconfirmed after transfer_receipt_days raises a REVIEW.
Opening balance (OPB)	At onboarding: counted quantities with batch data and cost, signed by the entity and the Federation onboarding officer (two signatures as state transitions); posts OPENING_BALANCE movements; the only way stock enters without a GRN or repack (R-02).

3.8 Loss

Aspect	Design
Categories	loss_category: DAMAGED_IN_TRANSIT, DAMAGED_IN_STORE, EXPIRED, THEFT, SHRINKAGE_UNEXPLAINED, STAFF_CONSUMPTION, SAMPLES, DONATION, OTHER (F-01). Each write-off line carries one; reporting never aggregates across them without showing the split.
Write-off (WOF)	Requester drafts (offline on the till allowed) with lines, category, batch, qty, photographs; a witness distinct from the requester confirms; an approver distinct from the requester approves within their value band (entity avg cost basis); posting creates WRITE_OFF movements. For a single-staff shop the MPCS approver acts as remote witness with photographs mandatory (doc 18 DR-7 adopted).
Attribution	The loss belongs to the entity owning the lot when the write-off is posted (ADR-03); DAMAGED_IN_TRANSIT on the seller's side is the seller's write-off of returned goods; on the buyer's side it is a claim (M4), not a write-off.
Expiry tasks	expiry_task created when a batch enters its window (SKU or tag expiry_warning_days): a count task for the holding location and a hand-off to M3's markdown rules; on expiry date, remaining quantity becomes a suggested write-off (category EXPIRED) that still needs the full lifecycle.
Tolerances and shrinkage	loss_tolerance(sku tag, pct or qty) below which count variances post without escalation (F-10); shrinkage percentage per shop is a projection of SHRINKAGE_UNEXPLAINED and negative-lot clearances over sales (F-09), visible to the Federation view-only.

3.9 Phase 2 as projections

Aspect	Design
Ageing and reorder	Ageing by lot received_at and reorder points from sales history are M8 projections over the movement ledger and M6 receipts; the ledger design needs nothing more; suggestions feed M4's requisition book (FR-ORD-010).

4. State machines

4.1 Count

From	To	Trigger / command	Guard	Permission	Audit / offline
–	SCHEDULED	ScheduleCount(scope, date)	entity scope; scope valid	inv.count.scheduled	COUNT_SCHEDULED
SCHEDULED	COUNTING	StartCount	location scope; expectation snapshot taken	shop.count.record	COUNT_STARTED / offline
COUNTING	SUBMITTED	SubmitCount(lines)	all scope lines counted or explicitly skipped; issuance (LOCATION series)	shop.count.record	COUNT_SUBMITTED / offline
SUBMITTED	VARIANCE_REVIEW	system: variances above tolerance exist	–	system	ADJ created DRAFT
SUBMITTED	CLOSED	system: no variance above tolerance	in-tolerance variances post automatically	system	COUNT_CLOSED
VARIANCE_REVIEW	CLOSED	ADJ approved or rejected	–	system	COUNT_CLOSED

4.2 Adjustment

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	system (from count) or CreateAdjustment(reason)	entity scope; lines with batches	inv.adjust.request	–
DRAFT	PENDING_APPROVAL	SubmitAdjustment	issuance (ENTITY series); value band computed	inv.adjust.request	ADJ_SUBMITTED
PENDING_APPROVAL	APPROVED	ApproveAdjustment	approver ≠ requester (SoD INSTANCE); approver's limit ≥ value; MFA	inv.adjust.approve (MFA)	ADJ_APPROVED
PENDING_APPROVAL	REJECTED	RejectAdjustment(reason)	reason	inv.adjust.approve	ADJ_REJECTED
APPROVED	POSTED	system: movements posted	same transaction as approval	system	ADJ_POSTED

4.3 Write-off

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	RequestWriteOff(lines, category, photos)	location scope; lot exists; qty ≤ on hand (or negative-clearing)	inv.writeoff.request	– / offline (draft)
DRAFT	REQUESTED	SubmitWriteOff	issuance (LOCATION series at a shop, ENTITY at warehouse); photographs referenced (mandatory when the location is single-staff or category is THEFT/SHRINKAGE)	inv.writeoff.request	WRITEOFF_REQUESTED / online
REQUESTED	WITNESSED	WitnessWriteOff	witness ≠ requester (SoD); attachments COMPLETE; single-staff shop: the approver's witnessing counts (remote)	inv.writeoff.witness	WRITEOFF_WITNESSED
WITNESSED	APPROVED	ApproveWriteOff	approver ≠ requester; value within approver's band (entity avg cost); MFA	inv.writeoff.approve (MFA)	WRITEOFF_APPROVED
REQUESTED / WITNESSED	REJECTED	RejectWriteOff(reason)	reason	inv.writeoff.approve	WRITEOFF_REJECTED
APPROVED	POSTED	system: WRITE_OFF movements	same transaction	system	WRITEOFF_POSTED

4.4 Repack

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	PlanRepack(recipe, input lots)	recipe ACTIVE; input lots GOOD with qty	inv.repack.execute	–

From	To	Trigger / command	Guard	Permission	Audit / offline
DRAFT	EXECUTED	ExecuteRepack(actual output qty)	issuance (LOCATION or ENTITY series); movements consume and produce; output batch registered (M2); variance recorded	inv.repack.execute	REPACK_EXECUTED / offline at shop
EXECUTED	REVERSED	ReverseRepack(reason)	output lot untouched (qty = produced; no movements against it); REVERSES-linked RPK	inv.repack.reverse	REPACK_REVERSED / online
EXECUTED	CLOSED	system: any movement against the output lot	–	system	–

4.5 Transfer

From	To	Trigger / command	Guard	Permission	Audit / offline
–	ISSUED	IssueTransfer (from M4 approval or direct)	same entity; source availability; issuance (LOCATION or ENTITY series); TRANSFER_OUT posted	inv.transfer.issue	TRANSFER_ISSUED
ISSUED	IN_TRANSIT	Dispatch	–	inv.transfer.issue	–
IN_TRANSIT	RECEIVED	ReceiveTransfer(actuals)	destination scope; TRANSFER_IN posted; variance → ADJ draft at destination	shop.transfer.receive	TRANSFER_RECEIVED / offline
ISSUED / IN_TRANSIT	CANCELLED	CancelTransfer(reason)	before receipt; TRANSFER_OUT reversed	inv.transfer.issue	TRANSFER_CANCELLED

4.6 Expiry task

From	To	Trigger / command	Guard	Permission	Audit / offline
–	OPEN	system (ExpiryScanJob)	batch enters window at a location	system	EXPIRY_TASK_OPENED
OPEN	COUNTED	linked count submitted	–	system	–
OPEN / COUNTED	CLOSED	lot exhausted, written off, or expiry passed with a suggested write-off created	–	system	EXPIRY_TASK_CLOSED

4.7 Opening balance

From	To	Trigger / command	Guard	Permission	Audit / offline
–	DRAFT	PrepareOpeningBalance(lines with batch data and cost)	location ONBOARDING; no prior movements at the location	inv.opening.prepare	–
DRAFT	SIGNED_ENTITY	SignOpeningBalance	entity officer; MFA	inv.opening.sign (entity, MFA)	OPB_SIGNED_ENTITY
SIGNED_ENTITY	POSTED	CountersignOpeningBalance	Federation onboarding officer; MFA; issuance (ENTITY series); OPENING_BALANCE movements	inv.opening.countersign (F, MFA)	OPB_POSTED

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Events
ApplyReceipt (internal, from grn.confirmed)	grn lines with batch, qty, damaged, cost	system	stock.moved.v1
ApplySale / ApplySaleReversal (internal, from receipt bundles)	receipt lines with batch, qty	system	stock.moved.v1; lot.negative.v1
ApplyReturn (internal, from claim.approved / drop.undelivered)	lines	system	stock.moved.v1
ScheduleCount / StartCount / SubmitCount / SkipCountLine	scope, date; lines (batch, condition, counted)	inv.count.schedule / shop.count.record	count.*.v1
CreateAdjustment / SubmitAdjustment / ApproveAdjustment / RejectAdjustment	lines, reason	inv.adjust.request / approve (MFA)	adjustment.*.v1; stock.moved.v1
RequestWriteOff / SubmitWriteOff / WitnessWriteOff / ApproveWriteOff / RejectWriteOff	lines, category, photos; reason	inv.writeoff.request / witness / approve (MFA)	writeoff.*.v1; stock.moved.v1
DefineRecipe / RetireRecipe	input/output skus and qty, expected loss	inv.recipe.manage	recipe.defined/retired.v1
PlanRepack / ExecuteRepack / ReverseRepack	recipe, input lots, actual output; reason	inv.repack.execute / reverse	repack.executed/reverse d.v1; stock.moved.v1
IssueTransfer / DispatchTransfer / ReceiveTransfer / CancelTransfer	source, destination, lines; actuals	inv.transfer.issue / shop.transfer.receive	transfer.*.v1; stock.moved.v1
PrepareOpeningBalance / SignOpeningBalance / CountersignOpeningBalance	lines with batch data and cost	inv.opening.* (MFA)	opening_balance.*.v1; stock.moved.v1
SetVarianceThreshold / SetLossTolerance / SetExpiryWindow	scope sku tag, values	inv.policy.set	inventory_policy.change d.v1
OverridePick	pick list line, lot, reason	whs.pick	pick.overridden.v1
AcknowledgeNegativeLot	lot, note	inv.count.schedule	lot.acknowledged.v1

5.2 Queries

Query	Returns	Scope rule
GetBalance(location, sku) / ListBalances(location, filter)	Lots by batch and condition with expiry, qty, FEFO rank; cost only for OWN	OWN (cost hidden at till), FEDERATION_VIEW
Availability(locations, skus)	Sellable quantities (Section 3.4)	OWN; PARTY (buyer of an ACTIVE relationship, indicative or live per seller config)
InStockBatches(locations, sku)	(batch, printed MRP, expiry, on hand) for M3 authoring checks	OWN (M3)
LotsConsumed(grn) / HasAnyLot(sku) / SkusWithLots(location)	Guards and assortment for M4 and M2	Internal
PickBatches(dn lines) / GetPickList(dn)	FEFO-suggested lots per line	OWN (seller)
EntityAverageCost(entity, sku)	avg_cost, qty	OWN, FEDERATION_VIEW
ListExpiring(location, window) / ListExpiryTasks	Batches and tasks	OWN
ListCounts / ListAdjustments / ListWriteOffs(scope, status)	Documents with lines and approval state	OWN; approver queues
LossSummary(scope, period, category) / ShrinkagePct(location, period)	For M8 and the Federation view	OWN, FEDERATION_VIEW
GetLotSnapshot(location, since)	Lots for the till (batch, MRP via M2, expiry, FEFO rank, qty)	Snapshot builder

5.3 Events published

Event	Payload	Consumers
stock.moved.v1	location, batch, condition, qty delta, type, cost at movement, document	M2 assortment (lots), snapshot (lot rows), M3 (in-stock MRPs), M8 projections, M10 (P2)
lot.negative.v1 / lot.acknowledged.v1	lot, qty, since, document that caused it	Exception queues (M1), M8
count.scheduled/started/submitted/closed.v1	task, location, scope, variances	Snapshot (count tasks to tills), notifications
adjustment.submitted/approved/rejected/posted.v1	document, value band, approver	Exception queues, M8
writeoff.requested/witnessed/approved/rejected/posted.v1	document, category, value, witness, approver	Exception queues, M8 (loss by category), notifications
repack.executed/reversed.v1	document, recipe, input lots, output batch, variance	M2 (output batch registration), M8
transfer.issued/dispatched/received/cancelled.v1	document, locations, lines	M4 (write-back of xfr id), snapshot (expected transfer lines at destination), M8
opening_balance.posted.v1	location, lines	M8; M1 (location activation gate may require it: DR-6)
batch.expiring.v1 (relayed as expiry_task.opened.v1)	batch, location, days to expiry, qty	M3 markdown rules, snapshot, notifications
inventory_policy.changed.v1	scope, thresholds	Snapshot (tolerances for the till count screen)

6. Key flows

Flow	Sequence	Failure paths
6.1 Receipt into stock	M4 grn.confirmed.v1 → per line: GOOD lot (received – damaged) and DAMAGED lot (damaged) at the line's unit cost; RECEIPT movements; entity average recomputed; M2 assortment adds the SKU for the location; lot snapshot delta	A GRN reversal (grn.reversed.v1) posts negating movements only if lots are unconsumed (M4 already guarded via LotsConsumed)
6.2 Sale deduction and oversell	Receipt bundle ingested (M6) → per line SALE movement against the resolved batch's lot → lot goes to –3 because two tills sold the same units offline → lot.negative.v1 → exception queue; availability excludes it → next count corrects; the cleared negative is reported as SHRINKAGE_UNEXPLAINED or as an over-count depending on the direction	A sale against a batch with no lot at the location (till cache stale) creates the lot at zero minus qty with cost = entity average and a REVIEW
6.3 Scheduled count with variance	MPCS schedules a monthly count of tag dairy for shop 7 → the task reaches the till in the snapshot → shop-in-charge counts offline; expectation snapshot taken at start; sales during the count adjust the expectation → submit (LOCATION series) → variances above tolerance form an ADJ draft → MPCS accountant approves within band (MFA) → COUNT_ADJUST movements	Variance above the accountant's band escalates to the chairman's band; a count on a location with an in-transit transfer warns
6.4 Write-off at a single-staff shop	Shop-in-charge finds six expired packs → drafts WOF offline with photos and category EXPIRED → submits when online (LOCATION series) → MPCS accountant, as remote witness, reviews photos and witnesses (attachments COMPLETE) → the same accountant cannot approve if they are also the requester (never the case here) → approves within band (MFA) → WRITE_OFF movements; loss attributed to the MPCS	Category THEFT without photos refused; approver band exceeded → routed to the next band; requester = approver refused by SoD
6.5 Repack and wrong-recipe correction (doc 13 scenario 3)	Warehouse repacks 50 kg bulk into 1 kg packs by recipe R-012 (expected 49 after loss) → actual 49 → REPACK_CONSUME 50 kg, REPACK_PRODUCE 49 EA at cost 9,500/49; output batch registered (M2) → discovers it should have been 5 kg packs → ReverseRepack (output untouched) restores the input lot → executes recipe R-013	Reversal after 12 packs were sold refused; correction then requires a write-off or a second recipe consuming the 1 kg packs as input
6.6 Transfer between two shops	M4 transfer_request.approved.v1 → IssueTransfer at shop B (TRANSFER_OUT at cost) → dispatched → shop A receives actuals on its till (TRANSFER_IN) → a short receipt raises an ADJ draft at A; the MPCS decides	Unreceived after transfer_receipt_days → REVIEW
6.7 Return to seller on an approved claim	M4 claim.approved.v1 with return_required → RETURN_TO_SELLER movement from the buyer's lot (DAMAGED or GOOD as the claim states) → the seller's receipt of the return is a GRN on the seller's side citing the claim	–
6.8 Expiry window	ExpiryScanJob finds batch B2411A entering its 30-day window at shop 7 → expiry task, count task, and expiry_task.opened.v1 → M3's markdown rule applies at the till from that day → on the expiry date the remainder becomes a suggested WOF (EXPIRED) awaiting the full lifecycle	–
6.9 Opening balance at onboarding	Federation onboarding officer and the MPCS count shop 7 with batch data and costs → PrepareOpeningBalance → MPCS officer signs (MFA) → Federation countersigns (MFA) → OPENING_BALANCE movements; lots and entity averages seeded; the location may activate (M1 gate, DR-6)	A second OPB for a location with movements refused

7. Configuration

Item	Default	Scope	Who may change
inventory.variance_threshold bands (sku tag): tolerance, band1 limit, band2 limit	tolerance 1% or 2 units; band1 Rs 25,000; band2 Rs 250,000; above: chairman	Entity	inv.policy.set
inventory.loss_tolerance (sku tag)	1% or 2 units	Entity	inv.policy.set
inventory.expiry_warning_days (sku tag)	30 (dairy 7, fresh 3)	Entity; Federation defaults by governed tag	inv.policy.set
inventory.writeoff_bands (value limits per approver role)	as role limits (M1)	Entity	gov.role.manage

Item	Default	Scope	Who may change
inventory.writeoff_photos_required	always for THEFT, SHRINKAGE, single-staff locations; otherwise recommended	Entity	inv.policy.set
inventory.remote_witness_allowed	true (single-staff shops)	Federation	inv.policy.set (F)
inventory.count_frequency (scope)	monthly full; weekly cycle by tag	Entity	inv.count.schedule
inventory.transfer_receipt_days	3	Entity	inv.policy.set
inventory.repack_reversal_requires_untouched_output	true (not configurable)	–	–
inventory.availability_excludes_negative	true (not configurable)	–	–

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
Stock book (position)	Stock book	W / T	Stores, shop	By SKU and batch; expiry badges; negative lots flagged; cost hidden at the till
Stock card (movement ledger)	Stock card	W	Stores, auditor	Per SKU with document links
Stocktake sheet (count)	Stocktake sheet	T (shop mode) / W	Shop, stores	Lines to count with thumbnails; enter counted; variance shown after entry; skip with reason; submit offline
Adjustment voucher	Adjustment voucher	W	Approver	Value band, approver limit, MFA
Damage/expiry register (write-off)	Damage register	T (shop mode) / W	Requester, witness, approver	Category, batch, qty, photos; witness and approval steps visible; SoD enforced visibly
Production sheet (repack)	Production sheet	W / T	Stores	Recipe, input lot, actual yield, variance; reversal while eligible
Transfer note	Transfer note	W / T	MPCS, shops	Issue at source; receive at destination offline
Pick list	Pick list	W	Warehouse	FEFO suggestions per DN line; override with reason
Expiry list	Expiry list	W / T	Supervisor	Batches in window with markdown rule (M3) and suggested write-offs
Opening stock sheet	Opening stock sheet	W	Onboarding officer, entity officer	Two signatures with step-up
Loss and shrinkage	–	W	Entity, Federation view	By category and shop; tolerances

9. Non-functional design

Aspect	Design
9.1 Capacity	Movements are the highest-volume table after receipt lines: about 15 million a day at the horizon (five lines per receipt). Partitioned monthly by <code>received_at</code> ; the lot table is small (locations × live batches). Balance updates are single-row increments in the ingestion transaction; the entity average update is one row per (entity, sku) per receipt-type movement.
9.2 Offline	Count entry, write-off draft, repack at a shop and transfer receipt are till-local documents in the LOCATION series (primary till) and arrive as bundles; the till deducts sales against its lot cache immediately and the ledger at central is authoritative after ingestion.
9.3 Consistency	Lot balance = sum of movements (B-14) enforced by the transaction pattern and verified nightly; negative lots are visible, never hidden; the reconciliation job raises, never fixes.
9.4 Controls	Write-off is the highest-fraud-risk lifecycle: SoD at INSTANCE level, witness rule, photographs, value bands, MFA on approval, every step audited; adjustments the same minus the witness; every negative lot and every tolerance-cleared variance lands on the exception report.
9.5 Visibility	Costs and margins are OWN and Federation view-only (M-04); the buyer of a relationship never sees the seller's lots except through availability; shops see only their own location (M-05).
9.6 Phase 2 readiness	Ageing, reorder points and ABC cycle counting are projections over this ledger; nothing in the phase-1 schema changes for them.

10. Test strategy

Layer	What is tested
Unit	Every guard of Section 4; weighted-average arithmetic to the cent through the doc 13 scenarios; FEFO ranking; availability formula; variance bands; tolerance evaluation; repack cost derivation and yield variance; SoD and witness rules
Contract	Every command and query; cost hidden in till-scoped balance reads; Availability for PARTY indicative vs live
State machine	All seven machines including offline transitions ingested as bundles
Property-based	Lot balance equals movement sum under random interleavings including reversals and negatives; entity average never negative and equals the recomputed-from-scratch value; a reversal restores the input lot exactly; availability never counts negative lots
Scenario	Doc 13 scenarios 3 and 4 (M5 parts); flows 6.1–6.9; oversell from two till simulators selling the same last units
Reconciliation	Injected drift is found and reported; nothing auto-fixed
Performance	Ingestion of 1,000 receipt bundles per second sustains balance updates; nightly reconcile of 10,000 locations under 30 minutes; expiry scan under 10 minutes

11. Decision requests

#	Point	Proposed	Decider
DR-1	Availability definition	On hand (non-negative GOOD lots) minus issued-undispatched DN quantities; allocation does not reserve	Federation with distributor input
DR-2	Damaged units as a separate lot condition	Adopt; sellable only via markdown or written off	Architecture review
DR-3	Write-off value basis for bands	Entity average cost (not retail)	Federation accounts
DR-4	Remote witness for single-staff shops	Adopt (doc 18 DR-7); photos mandatory	Federation admins
DR-5	Oversell against a batch with no lot at the location	Create the lot negative with a REVIEW rather than reject	Architecture review
DR-6	Opening balance as a prerequisite for location activation	Yes for SHOP and WAREHOUSE (M1 gate)	Federation admins
DR-7	New tables (CR-25-1 to doc 18): <code>entity_sku_cost</code> , <code>count_task</code> , <code>expiry_task</code> , <code>pick_list</code> , <code>loss_category</code> , <code>variance_threshold</code> , <code>loss_tolerance</code> , document extensions	Adopt	Architecture review
DR-8	Count-while-trading expectation adjustment	Apply movements during the count to the expectation (E-06)	Architecture review, till track

12. Traceability

Section	Identifiers
3.1–3.3	FR-INV-010, 020, 060; E-01 to E-03, E-09; ADR-03, ADR-38; doc 18 B-I3 to B-I5
3.4	FR-INV-030; E-04; C-12; doc 24 allocation
3.5	FR-INV-040, 050; E-05 to E-07; F-10
3.6	FR-INV-080; B-06; ADR-07; doc 13 scenario 3
3.7	FR-INV-070; C-10, C-11; R-02
3.8	FR-LOSS-010 to 060; F-01 to F-11; N-07; doc 18 DR-7
3.9	FR-INV-090, 100; E-08, E-11
4	Doc 03 §9–11; H-02/H-03
9	M-04, M-05; NFR-PRF; NFR-AUD

End of document. Draft for design review; the contract freezes when M6 acknowledges it, targeted end of Stage 2 week 6.
Implementation guide: 25A.